# Bond Sphere: Diatomic Molecules as $S^3$ and their Natural Sturmian Basis*

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A diatomic molecule is geometrically a single three-sphere $S^3$ with two weighted poles—the *bond sphere*—in which the internuclear distance $R$ encodes as a polar angle $\gamma$ via $\cos\gamma = (p_0^2 - p_R^2)/(p_0^2 + p_R^2)$ and all cross-atom matrix elements are $SO(4)$ Wigner $D$-matrix elements evaluated at the bond rotation $e^{i\gamma A_y}$. *Positive results:* The bond sphere geometry is consistent, with the $\gamma(R)$ encoding, heteronuclear egg metric, Runge–Lenz alignment, and kinetic membrane all following from $SO(4)$ representation theory. The $D$-matrix enforces selection rules for $\sigma$-bonds: $D^2_{10,10}(\gamma) = 1$ and $D^2_{00,10}(\gamma) = 0$ for all $\gamma$. *Negative theorem:* In the molecular Sturmian basis with shared momentum scale $p_0$, $H_{ij} \propto S_{ij}$; eigenvalues are independent of $R$. Binding requires the $R$-dependent prolate spheroidal spectrum $\beta_k(R)$, reintroducing continuous geometry. The resolution—discretizing the Laplace–Beltrami operator in prolate spheroidal coordinates—is developed in the companion Paper 11 [? ].

## I. INTRODUCTION

In the GeoVac framework [? ? ? ], quantum numbers $(n, l, m)$ are not abstract labels but geometric coordinates on a three-sphere $S^3$. The principal quantum number $n$ indexes latitude shells, each containing $n^2$ nodes; the orbital angular momentum $l$ and magnetic quantum number $m$ locate a node within its shell. The graph Laplacian over these nodes is mathematically equivalent to the Laplace–Beltrami operator on the unit $S^3$, and the continuous Schrödinger equation emerges as its flat-space projection via Fock's 1935 stereographic map [? ? ].

This geometric picture is complete for single atoms. For molecules, the current implementation stitches together two separate $S^3$ manifolds—one per atom—using approximate cross terms: Mulliken overlap for cross-nuclear attraction, Fourier convolution for direct Coulomb integrals, and STO-overlap bridges for inter-center hopping [? ]. The result is basis set superposition error (BSSE), approximate integrals, and no natural coordinate system for the bonding region.

This paper proposes the resolution: the *bond sphere*. A diatomic molecule is described by a single $S^3$ with two weighted poles, one per nucleus. The atomic $S^3$ manifolds are the $R \to \infty$ limit. Cross-atom matrix elements that are currently approximated become exact $SO(4)$ Wigner $D$-matrix elements. The natural basis for this geometry is the Sturmian basis, in which all functions share a single momentum parameter $p_0$ and therefore live on the same $S^3$ by construction. The $\beta_k$ eigenvalues that encode nuclear attraction are determined by the two-center prolate spheroidal eigenvalue problem—whose coordinate surfaces are precisely the confocal ellipsoids of the bond sphere metric.

The paper is organized as follows. Sections **??** and **??** establish the single-atom and diatomic bond sphere geometry. Section **??** gives the Runge–Lenz alignment argument for why the bond rotation is generated by $A_y$. Section **??** treats heteronuclear asymmetry—the egg. Section **??** identifies the kinetic membrane as the Fiedler vector zero-crossing. Section **??** introduces the Sturmian basis. Section **??** develops the $\beta_k$ eigenvalue problem via prolate spheroidal separation. Section **??** proves why atom-centered approximations fail. Section **??** states the uniform diagonal theorem. Section **??** proves the Sturmian structural theorem: the Hamiltonian is proportional to the overlap matrix, making eigenvalues geometry-independent. Section **??** derives the $SO(4)$ selection rules for $\sigma$-bond formation. Section **??** concludes.

## II. WHAT IS KNOWN VS. WHAT IS NEW

### A. Known Results (Prior Work)

- *Shibuya and Wulfman (1965)* [? ]: The two-center Coulomb problem in momentum space maps to a single $S^3$ with two poles. Sturmian basis functions on this sphere provide a complete, orthogonal expansion.
- *Aquilanti and Cavalli (1980s–1990s)* [? ? ]: Hyperspherical harmonic approach to molecular problems, including systematic use of $SO(4)$ Wigner $D$-matrices and the relationship between internuclear distance and angular pole separation on $S^3$.
- *Avery (2000s)* [? ]: Extended the Sturmian basis to general molecular systems with practical computational methods.
- $SO(4)$ *representation theory* [? ? ]: The Wigner $D$-matrix elements of $SO(4)$ in the $(n, l, m)$ basis, generated by angular momentum $\mathbf{L}$ and the Runge–Lenz vector $\mathbf{A}$.

* Paper 8–9 in the Geometric Vacuum series

- *Sturmian basis functions*: Established by Shull and Löwdin [? ], Rotenberg [? ], and Aquilanti–Avery [? ? ]. Orthogonality under $1/r$ weight and completeness for the Coulomb problem.

### B. New Contributions

1. **Bond sphere as a GeoVac lattice object.** The bond sphere is the natural molecular generalization of the single-atom $S^3$ lattice (Papers 0, 1, 7), with within-block edges generated by $T_\pm, L_\pm$ and cross-block edges weighted by $D$-matrix elements at $g(\gamma(R))$.

2. **Runge–Lenz alignment argument** (Sec. **??**). The poles are forced onto the bond axis by dynamics: the Runge–Lenz vector of each atom orients toward the other nucleus.

3. **Heteronuclear egg geometry** (Sec. **??**). For $Z_A \neq Z_B$, the bond sphere metric is a prolate spheroid with the heavier nucleus at the fat end.

4. **Kinetic membrane** (Sec. **??**). The Fiedler vector zero-crossing identifies a discrete, combinatorial analog of Bader's bond critical point.

5. **Identification that the bond sphere requires the Sturmian basis** (Sec. **??**). The single-$S^3$ geometry is only consistent when all basis functions share one $p_0$.

6. **Proof that atom-centered Sturmian bases fail for heteronuclear molecules** (Sec. **??**), with the precise threshold $\beta > p_0/(2Z_B)$ for binding.

7. **Uniform diagonal theorem** (Sec. **??**). In the molecular Sturmian basis, $\varepsilon_k = -p_0^2/2$ for every orbital.

8. **Sturmian structural theorem** (Sec. **??**). In the molecular Sturmian basis with shared $p_0$, the Hamiltonian is proportional to the overlap matrix; eigenvalues are independent of the bond angle $\gamma$ and hence of $R$. Binding requires $\beta_k(R)$ from the full PDE.

9. $SO(4)$ **selection rules for $\sigma$-bonds** (Sec. **??**). $D^2_{10,10}(\gamma) = 1$ and $D^2_{00,10}(\gamma) = 0$ for all $\gamma$: the bond rotation preserves $m = 0$ within $l = 1$ and does not mix $s$ and $p$ states.

## III. THE SINGLE-ATOM CASE AS A SPECIAL LIMIT

The existing GeVac single-atom lattice is the $\gamma = 0$ limit of the bond sphere, so all results of Papers 0–7 are preserved.

### A. The $T_\pm, L_\pm$ Operators as $SO(4)$ Generators

The GeoVac paraboloid lattice is generated by four transition operators (Paper 7, Eqs. (5)–(6)):

$$L_\pm|n,l,m\rangle \propto |n,l,m\pm 1\rangle, \tag{1}$$
$$T_\pm|n,l,m\rangle \propto |n\pm 1,l,m\rangle. \tag{2}$$

The angular momentum operators $L_x, L_y, L_z$ satisfy $\mathfrak{su}(2)$:

$$[L_i, L_j] = i\epsilon_{ijk}L_k. \tag{3}$$

The scaled Runge–Lenz vector $\mathbf{A}$, generated by $T_\pm$ within each $n$-shell, satisfies [? ]:

$$[L_i, A_j] = i\epsilon_{ijk}A_k, \tag{4}$$
$$[A_i, A_j] = i\epsilon_{ijk}L_k. \tag{5}$$

The $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ structure is exhibited by

$$\mathbf{J}^{(\pm)} = \tfrac{1}{2}(\mathbf{L} \pm \mathbf{A}), \tag{6}$$

which satisfy two independent, commuting $\mathfrak{su}(2)$ algebras. For principal quantum number $n$, both carry spin $j = (n-1)/2$, giving the $(n-1)/2 \otimes (n-1)/2$ representation with Casimir

$$\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1, \tag{7}$$

recovering the $S^3$ eigenvalue spectrum $\lambda_n = -(n^2-1)$ of Paper 7.

### B. Covariance and the Single-Atom Limit

Because $T_\pm, L_\pm$ generate $SO(4)$, the graph Laplacian $L = D - A$ is covariant under $SO(4)$ transformations. The second atom can be introduced as an $SO(4)$-rotated copy, with the rotation angle determined by $R$. Cross-atom matrix elements are then $SO(4)$ Wigner $D$-matrix elements.

At $\gamma = 0$ (coincident poles): $g(0) = e$, the $D$-matrix reduces to Kronecker deltas, cross-block edges vanish, and the single-center lattice of Papers 0–7 is recovered. The 18 symbolic proofs, the dimensionless vacuum principle, and the universal constant $-1/16$ are unchanged.

## IV. THE BOND SPHERE CONSTRUCTION

### A. The Angle–Distance Relationship

The internuclear distance $R$ maps to a polar angle $\gamma$ between two poles on $S^3$. The momentum conjugate to $R$ defines

$$p_R = \frac{1}{R}, \tag{8}$$

and the angular separation is

$$\cos\gamma = \frac{p_0^2 - p_R^2}{p_0^2 + p_R^2}, \tag{9}$$

which is the fourth component $n_4$ of the Fock projection evaluated at $|\mathbf{p}| = p_R$. The sine follows as

$$\sin\gamma = \frac{2p_0\, p_R}{p_0^2 + p_R^2} = \Omega(p_R) \cdot p_R, \tag{10}$$

where $\Omega$ is the Fock conformal factor.

*a. Limiting behavior.* At $R \to \infty$: $\gamma \to 0$ (coincident poles, single-atom limit). At $R \to 0$: $\gamma \to \pi$ (antipodal poles, united-atom limit).

### B. Eigenfunctions as $SO(4)$ Wigner $D$-Matrix Elements

When a second pole is introduced at angle $\gamma$, the cross-center matrix elements are

$$\langle n', l', m' | U(g(\gamma)) | n, l, m\rangle = D^{(n)}_{(l'm'),(lm)}(g(\gamma)), \tag{11}$$

where $g(\gamma)$ is the $SO(4)$ rotation that maps one pole to the other. The current Mulliken/Fourier/bridge cross-atom terms are approximations to these exact $D$-matrix elements.

### C. The Bond-Sphere Lattice

*a. Nodes.* Each node is labeled $(n, l, m, \alpha)$ with $\alpha \in \{A, B\}$. Total spatial orbitals: $2N_{\rm sp}$.

*b. Within-block edges.* Generated by $T_\pm, L_\pm$ at each center, as in the single-atom lattice.

*c. Cross-block edges.* Weighted by $D$-matrix elements:

$$w_{(n'l'm',B),(nlm,A)} = D^{(n)}_{(l'm'),(lm)}(g(\gamma(R))). \tag{12}$$

*d. Hamiltonian structure.*

$$H_1 = \begin{pmatrix} H_A & V_{AB} \\ V_{AB}^\dagger & H_B \end{pmatrix}, \tag{13}$$

where $H_A, H_B$ are single-atom Hamiltonians and $V_{AB}$ has elements given by Eq. (**??**) modulated by nuclear charges and kinetic scale.

### D. The Opposite-Sign Rotation

The bond rotation is not a spatial rotation. Since $\mathbf{A} = \mathbf{J}^{(+)} - \mathbf{J}^{(-)}$, the rotation generated by $A_y$ acts as

$$e^{i\gamma A_y} = e^{i\gamma J_y^{(+)}}\, e^{-i\gamma J_y^{(-)}}, \tag{14}$$

where the $\mathbf{J}^{(+)}$ and $\mathbf{J}^{(-)}$ sectors rotate in opposite directions: $+\gamma$ and $-\gamma$. A pure spatial rotation $e^{i\alpha L_y}$ gives same-sign rotation $d^j(+\alpha) \otimes d^j(+\alpha)$, preserving radial structure. The bond rotation gives $d^j(+\gamma) \otimes d^j(-\gamma)$, mixing $s, p, d, \ldots$ orbitals within each $n$-shell. This $s$–$p$ hybridization is precisely what chemical bonding requires.

The $D$-matrix in the $(n, l, m)$ basis is

$$D^{(n)}_{(l'm'),(lm)}(\gamma) = \sum_{m^+,m^-} \sum_{m'^+,m'^-} C^n_{l'm';m'^+m'^-}\, C^n_{lm;m^+m^-} \times\, d^j_{m'^+,m^+}(+\gamma)\, d^j_{m'^-,m^-}(-\gamma), \tag{15}$$

where $j = (n-1)/2$ and $C^n_{lm;m^+m^-}$ are the $SO(4)$ Clebsch–Gordan coefficients coupling $|j, m^+\rangle \otimes |j, m^-\rangle \to |l, m\rangle$ with $m = m^+ + m^-$.

### E. The Antipodal Limit

At $\gamma = \pi$ ($R = 0$, united atom):

$$D^{(n)}(\pi)^2 = \mathbf{I}_{n^2}, \tag{16}$$

with eigenvalues $\pm 1$ and parity structure $|l, m\rangle \to (-1)^{n-1+l+|m|}\, |l, -m\rangle$—the Runge–Lenz parity, not spatial parity $(-1)^l$.

## V. THE RUNGE–LENZ ALIGNMENT ARGUMENT

The bond rotation $e^{i\gamma A_y}$ is generated by $A_y$, a component of the Runge–Lenz vector perpendicular to the bond axis. This choice is not a convention but a dynamical consequence.

For a hydrogen atom, the Runge–Lenz vector $\mathbf{A}$ points from the nucleus toward the perihelion of the classical orbit. In a diatomic molecule, the electrostatic field of the partner nucleus breaks the spherical symmetry of each atom's potential. The lowest-energy configuration aligns the Keplerian orbit so that its perihelion points toward the attracting partner—that is, $\mathbf{A}$ orients along the internuclear axis. With $\mathbf{A}$ aligned along $\hat{z}$ (the bond axis), the rotation that maps one pole to the other is generated by $A_y$ (or $A_x$), which is perpendicular to the bond.

This is the physical origin of $s$–$p$ hybridization in the bond sphere picture: the Runge–Lenz vector aligns with the bond, and rotation about a perpendicular axis mixes orbital angular momentum states. The hybridization is not an ansatz imposed on the basis—it is a consequence of the two-center electrostatic potential acting on the dynamical symmetry generators of each atom.

## VI. HETERONUCLEAR GEOMETRY: THE EGG

For a homonuclear diatomic ($Z_A = Z_B$), the bond sphere has uniform metric and full $SO(4)$ symmetry.

For a heteronuclear diatomic ($Z_A \neq Z_B$), the geometry changes qualitatively.

### A. Metric Distortion

The Fock conformal factor $\Omega = 2p_0/(p^2+p_0^2)$ maps momentum space to $S^3$. When the two nuclei have unequal charges, the natural momentum scale differs between the two poles: center $A$ (charge $Z_A$) favors $p_0 \sim Z_A$, while center $B$ (charge $Z_B$) favors $p_0 \sim Z_B$. With a shared $p_0$ (as required by the single-$S^3$ construction), the conformal factor is distorted asymmetrically about the two poles.

The result is a bond sphere whose intrinsic metric is not uniform. The heavier nucleus ($Z_A > Z_B$) occupies a larger solid angle on the sphere—a "fat end"—while the lighter nucleus occupies a smaller solid angle—a "narrow end." The overall shape is a prolate spheroid, with the heavier nucleus at the fat end and the lighter nucleus at the narrow end. We call this the *egg*.

### B. Bond Polarity as Egg Asymmetry

The degree of asymmetry is set by the ratio $Z_A/Z_B$. For LiH ($Z_A/Z_B = 3$), the egg is noticeably asymmetric. For $H_2$ ($Z_A/Z_B = 1$), the egg is a round sphere. This asymmetry is the geometric encoding of bond polarity: a polar bond is an asymmetric egg, and a nonpolar bond is a symmetric sphere.

### C. Topology vs. Metric

The node count per shell ($n^2$ nodes at latitude $n$ from each pole) is fixed by the Paper 0 packing topology. The egg deformation changes the *node weights*—the effective nuclear attraction felt by each node—but not the *graph topology*. The adjacency structure ($T_\pm, L_\pm$ edges) is unchanged; only the diagonal elements of the Hamiltonian (the node weights) and the cross-block edge weights (the $D$-matrix elements) are affected by the asymmetry.

### D. Symmetry Breaking: $SO(4) \to SO(2)$

The unequal pole weights break $SO(4)$ to the axial $SO(2)$ subgroup preserving the bond axis. The magnetic quantum number $m$ remains a good quantum number (generating $\sigma, \pi, \delta, \ldots$ labels), while the off-axis generators $L_\pm$ are no longer symmetries:

$$[\Delta_{\text{asym}}, L_z] = 0, \tag{17}$$

$$[\Delta_{\text{asym}}, L_\pm] \neq 0 \quad \text{for } Z_A \neq Z_B. \tag{18}$$

### E. Prolate Spheroidal Coordinates

The natural coordinate system for the egg is prolate spheroidal coordinates $(\xi, \eta, \varphi)$ with foci at the two nuclear poles. The coordinate surfaces are confocal ellipsoids ($\xi = \text{const}$) and hyperboloids ($\eta = \text{const}$). These are precisely the level sets of the distorted bond sphere metric: the ellipsoids trace curves of constant "distance from both poles simultaneously," while the hyperboloids trace curves of constant "distance ratio between the two poles." This coordinate system appears again in Sec. **??** as the natural domain for the $\beta_k$ eigenvalue problem.

## VII. THE KINETIC MEMBRANE

Between the two poles of the bond sphere lies a region where the edge weights from the two centers are comparable in magnitude. This is the *kinetic membrane*—not a separate geometric object, but an emergent feature of the edge weight pattern.

### A. Identification via the Fiedler Vector

In the molecular graph, the kinetic membrane is identified by the *Fiedler vector*—the eigenvector corresponding to the second smallest eigenvalue of the graph Laplacian. The Fiedler vector partitions the graph into two regions (positive and negative components), and the membrane is where the Fiedler vector crosses zero.

For a homonuclear diatomic, the Fiedler vector zero-crossing lies at the midpoint between the two poles, by symmetry. For a heteronuclear molecule, the zero-crossing shifts toward the lighter nucleus. In LiH, the heavier Li pole ($Z = 3$) dominates a larger volume of the bond sphere, pushing the membrane toward the H end.

### B. Connection to Bader's Bond Critical Point

The Fiedler vector zero-crossing is the discrete, combinatorial analog of Bader's bond critical point [**?** ] in the Quantum Theory of Atoms in Molecules (QTAIM). Bader's bond critical point is defined as the point where $\nabla\rho = 0$ along the bond path, where $\rho$ is the electron density. The GeoVac graph gives the same object without invoking the electron density: the membrane position is determined purely by the graph Laplacian spectrum.

For heteronuclear molecules, both the Bader critical point and the Fiedler zero-crossing shift toward the electropositive (lighter) atom. The membrane is thinner (steeper Fiedler gradient) for strongly polar bonds and thicker (shallower gradient) for nonpolar bonds.

## VIII. THE STURMIAN BASIS

The bond sphere hypothesis requires all basis functions to live on a single $S^3$ parameterized by one momentum scale $p_0$. The hydrogenic eigenstate basis uses $p_0(n) = Z/n$—a different sphere per shell—which is geometrically inconsistent with the single-$S^3$ construction. The Sturmian basis resolves this.

### A. Definition

The Sturmian basis functions $S_{nlm}$ are solutions to the scaled Coulomb problem:

$$\left(-\tfrac{1}{2}\nabla^2 - \frac{\beta_n Z}{r}\right) S_{nlm} = E_0\, S_{nlm}, \tag{19}$$

where $E_0 = -p_0^2/2$ is a fixed reference energy shared by all functions, and

$$\beta_n = \frac{p_0\, n}{Z} \tag{20}$$

is the shell-dependent scaling that enforces this.

The key property: all Sturmian functions share the same $E_0$ regardless of $n$. Under Fock's projection, all functions live on the *same* $S^3$ of radius $p_0$.

### B. Verification

Substituting effective charge $\tilde{Z} = \beta_n Z$ into the hydrogenic eigenvalue:

$$E_0 = -\frac{(\beta_n Z)^2}{2n^2} = -\frac{(p_0 n/Z)^2 Z^2}{2n^2} = -\frac{p_0^2}{2}, \tag{21}$$

independent of $n$, as required.

### C. Orthogonality under $1/r$ Weight

Sturmian functions satisfy

$$\left\langle S_{n'l'm'} \left| \frac{1}{r} \right| S_{nlm} \right\rangle = \frac{p_0}{n}\, \delta_{nn'}\, \delta_{ll'}\, \delta_{mm'}. \tag{22}$$

The $1/r$-weighted completeness precisely matches the Coulomb kernel in cross-center integrals, making the two-center problem exact.

### D. Geometric Inconsistency of the Hydrogenic Basis

The hydrogenic eigenstate $\psi_{nlm}$ has $p_0(n) = Z/n$. For the cross-center matrix element $\langle \psi^A_{n'l'm'} | -Z_B/r_B | \psi^A_{nlm} \rangle$ with $n' \neq n$, the bra lives on $S^3$ with radius $Z/n'$ and the ket on $S^3$ with radius $Z/n$. The integral involves two different Fock projections—a convolution over mismatched spheres. This is the origin of the $\sin\gamma$ form factor in the current SW module: a lowest-order correction for this geometric mismatch.

In the Sturmian basis, the mismatch is zero by construction.

### E. SW Integrals Are Exact in the Sturmian Basis

This is the central result. In the Sturmian basis:

$$\boxed{\left\langle S^A_{n'l'm'} \left| -\frac{Z_B}{r_B} \right| S^A_{nlm} \right\rangle = -\frac{Z_B}{p_0}\, D^{(n)}_{(l'm'),(lm)}(\gamma),} \tag{23}$$

exactly, with no form factor.

The derivation: the cross-center Coulomb potential is related to the same-center potential by an $SO(4)$ rotation (Eq. (**??**)). Using Sturmian orthogonality (Eq. (**??**)), the $1/r_A$ matrix element collapses to a Kronecker delta, and the sum over intermediate states reduces to a single $D$-matrix element.

When the same calculation is attempted with hydrogenic eigenstates of different $p_0$ values, the orthogonality does not apply, and one obtains the mismatch integral that, expanded to leading order, gives the $\sin\gamma$ form factor.

### F. The Self-Consistency Condition

The momentum parameter $p_0$ is determined by

$$p_0 = \sqrt{-2E_{\text{mol}}}, \tag{24}$$

where $E_{\text{mol}}$ is the ground-state eigenvalue of $H(p_0)$. This is a self-consistent fixed point, whose existence follows from the intermediate value theorem: $f(p_0) = \sqrt{-2E_{\text{mol}}(p_0)}$ maps $\mathbb{R}^+$ to $\mathbb{R}^+$, with $f(p_0) > p_0$ for small $p_0$ (diffuse basis, weakly negative energy) and $f(p_0) < p_0$ for large $p_0$ (compact basis, bounded energy).

### G. $\gamma$ Encodes Both Geometry and Energy

At self-consistency:

$$\gamma = \arccos\left(\frac{p_0^2 - p_R^2}{p_0^2 + p_R^2}\right), \tag{25}$$

with $p_0 = \sqrt{-2E_{\text{mol}}}$. Both derivatives $\partial\gamma/\partial R \neq 0$ and $\partial\gamma/\partial E_{\text{mol}} \neq 0$, confirming that $\gamma$ carries information about both bond length and molecular energy simultaneously.

### H. Backward Compatibility with Papers 0–7

In the limit $p_0 \to Z/n$ per shell: $\beta_n \to 1$, the Sturmian equation becomes the ordinary hydrogenic equation, and the atomic eigenvalue $-Z^2/(2n^2)$ is recovered. The self-consistency loop has no single-atom analog because $E_n = -Z^2/(2n^2)$ already determines $p_0(n) = Z/n$. All single-atom results are the $n$-resolved limit of the Sturmian construction.

## IX. THE $\beta_k$ EIGENVALUE PROBLEM

For a single-center Sturmian, $\beta_n = p_0 n/Z$ is determined by the nuclear charge alone. For a two-center molecular Sturmian, $\beta_k$ is the eigenvalue of the prolate spheroidal equation with foci at the two nuclei.

### A. Prolate Spheroidal Separation

The Sturmian equation with the two-center molecular potential $V = -Z_A/r_A - Z_B/r_B$ separates in prolate spheroidal coordinates $(\xi, \eta, \varphi)$ with foci at the nuclear positions. The separation yields an angular eigenvalue problem (in $\eta$) and a radial eigenvalue problem (in $\xi$), coupled by a separation constant $A$.

The angular equation at fixed magnetic quantum number $m$ and spheroidal parameter $c = p_0 R/2$ is

$$\frac{d}{d\eta}\left[(1-\eta^2)\frac{dS}{d\eta}\right] + \left(A_{m,n_{\rm sph}} - c^2\eta^2 - \frac{m^2}{1-\eta^2}\right) S = 0, \tag{26}$$

whose eigenvalues $A_{m,n_{\rm sph}}(c,b)$ depend on $c$ and the charge asymmetry parameter $b = (Z_A - Z_B)R/2$.

The radial equation is

$$\frac{d}{d\xi}\left[(\xi^2-1)\frac{dR}{d\xi}\right] + \left(-A + c^2\xi^2 + 2a\xi - \frac{m^2}{\xi^2-1}\right) R = 0, \tag{27}$$

where $a = \beta_k(Z_A + Z_B)R/2$ encodes the molecular Sturmian eigenvalue $\beta_k$.

### B. The Matching Condition

The molecular $\beta_k$ for each orbital $(m, n_{\rm sph}, n_{\rm rad})$ is determined by the matching condition: the angular separation constant $A_{m,n_{\rm sph}}(c,b)$ and the radial eigenvalue must be consistent. Specifically, $\beta_k$ is found by root-finding (brentq) on the condition

$$A_{\rm ang}(c,b) + L_{\rm rad}(c, a(\beta_k)) = 0, \tag{28}$$

where $L_{\rm rad}$ is the top radial eigenvalue at the given parameters.

### C. Key Result

**Theorem 1** (Molecular Sturmian $\beta_k$ for LiH)**.** *For LiH at $R = 3.015$ bohr with $p_0 = \sqrt{-2E_{mol}}$, the molecular Sturmian $\beta_k$ values computed via prolate spheroidal separation are:*

| *Orbital* | $\beta_k$ | *Character* |
|---|---|---|
| $1\sigma$ | *1.033* | *Li 1s core* |
| $2\sigma$ | *1.950* | *Li-H bonding* |
| $1\pi$ | *1.976* | *Li-H $\pi$* |

*All $n = 3$ orbitals have $\beta_k > 2.3$. The $H_2^+$ ground-state energy computed by this method is $E = -1.096$ Ha (exact: $-1.103$ Ha, 0.6% error).*

### D. Why Prolate Spheroidal Coordinates Are Natural

The separability of the Sturmian equation in prolate spheroidal coordinates is not a mathematical accident. The coordinate foci are the two nuclear poles, and the coordinate surfaces are the confocal ellipsoids that are the level sets of the distorted bond sphere metric (Sec. **??**). The coordinates respect both poles simultaneously—which is precisely the requirement for a basis that lives on the bond sphere rather than on a single-center sphere.

## X. WHY ATOM-CENTERED BASES FAIL FOR HETERONUCLEAR MOLECULES

**Theorem 2** (Atom-centered Sturmian failure)**.** *For a heteronuclear diatomic with $Z_A/Z_B = k > 1$, there is no atom-centered Sturmian basis with shared $p_0$ that simultaneously binds both centers.*

*Proof sketch.* The shared $p_0$ is determined by the deeper well ($Z_A$). At this $p_0$, the one-electron diagonal for the lightest orbital of center $B$ is

$$\varepsilon_1(p_0) = \frac{p_0^2}{2} - Z_B\, p_0. \tag{29}$$

This is positive (unbound) when $p_0 > 2Z_B$. For the self-consistent $p_0 \approx \sqrt{-2E_A}$ of the heavier atom with $Z_A \gg Z_B$, this condition is generically satisfied.

The cross-nuclear correction from $Z_A$ on the $B$-center $1s$ function, evaluated in the atom-centered Sturmian basis, is $\Delta\varepsilon = -(Z_A/p_0)D^{(1)}(\gamma)$. For $n = 1$, $D^{(1)} = 1$, so $\Delta\varepsilon = -Z_A/p_0$. The total diagonal becomes

$$\varepsilon_1^{\rm tot} = \frac{p_0^2}{2} - Z_B p_0 - \frac{Z_A}{p_0}. \tag{30}$$

For binding: $\varepsilon_1^{\rm tot} < 0$, i.e., $\beta_{\rm eff} > p_0/(2Z_B)$. But the atom-centered $\beta_1 = p_0/Z_B$ gives $\varepsilon_1 = p_0^2/2 - Z_B p_0$ plus the cross-nuclear $-Z_A/p_0$. At $p_0 = 3.168$ (LiH): $\varepsilon_1^{\rm tot} = 5.02 - 3.17 - 0.95 = +0.85$ Ha—still positive.

The cross-nuclear correction is insufficient because the atom-centered function has the wrong spatial extent: it was defined against $Z_B$ alone and is too compact near the $B$ center to capture the full $A$-$B$ cross-nuclear attraction. The correct integral requires the *molecular* Sturmian function—the two-center prolate spheroidal eigenfunction—which has the correct spatial extent near both centers. □

**Corollary 1.** *For LiH ($Z_A/Z_B = 3$): the binding threshold is $\beta > p_0/(2Z_B) = 1.581$. The $2\sigma$ molecular Sturmian achieves $\beta = 1.950$ (Theorem **??**), but no atom-centered approximation reaches this threshold.*

This argument is independent of the number of electrons and applies to any heteronuclear diatomic with $Z_A/Z_B$ sufficiently different from 1. The diagnostic arc v0.9.20–v0.9.30 (documented in the companion paper [**?** ]) exhaustively verifies this theorem computationally: every atom-centered variant fails for LiH by the mechanism described above.

## XI. THE UNIFORM DIAGONAL THEOREM

**Theorem 3** (Uniform diagonal)**.** *In the molecular Sturmian basis with shared $p_0$, the diagonal of the one-electron Hamiltonian is*

$$\varepsilon_k = -\frac{p_0^2}{2} \tag{31}$$

*for every orbital $k$, regardless of quantum numbers.*

*Proof.* Each molecular Sturmian $S_k$ satisfies $(-\frac{1}{2}\nabla^2 - \beta_k V_{\mathrm{mol}})S_k = E_0 S_k$ with $E_0 = -p_0^2/2$. The diagonal matrix element of the full one-electron Hamiltonian $\hat{h}_1 = -\frac{1}{2}\nabla^2 + V_{\mathrm{mol}}$ is

$$\langle S_k|\hat{h}_1|S_k\rangle = E_0 + (1-\beta_k^{-1})\,\langle S_k|V_{\mathrm{mol}}|S_k\rangle. \tag{32}$$

In the exact molecular Sturmian basis with proper normalization, the $V_{\mathrm{mol}}$ expectation value is $-p_0^2/\beta_k$, giving

$$\varepsilon_k = -\frac{p_0^2}{2} - \frac{p_0^2}{\beta_k} + \frac{p_0^2}{\beta_k} = -\frac{p_0^2}{2}. \tag{33}$$

□

*a. Physical interpretation.* The uniform diagonal means that no orbital has a different diagonal energy. The entire information content of the nuclear attraction is in the *off-diagonal* $\beta_k$ structure and the $D$-matrix coupling. The asymmetry between the two centers enters only through the $\beta_k$ eigenvalues (which differ for $\sigma$ vs. $\pi$ MOs and depend on $Z_A, Z_B, R$) and through the $D$-matrix rotation angle $\gamma$.

*b. Consistency with the $\beta_k$ discussion.* The $\beta_k$ eigenvalues (Sec. **??**) parametrize the nuclear attraction: orbital $k$ has effective charge $\beta_k Z$ in the Sturmian equation. This information enters the FCI Hamiltonian through the off-diagonal matrix elements (the $D$-matrix coupling weighted by $\beta_k$-dependent factors), not through the diagonal. The diagonal is uniformly $-p_0^2/2$, and all orbital differentiation is off-diagonal. This is the correct architecture for a bond sphere FCI: the basis is adapted to the molecular potential, and the nuclear charges appear in the coupling structure, not in the orbital energies.

## XII. GEOMETRIC LIMITS OF THE STURMIAN BOND SPHERE

The computational diagnostic arc v0.9.20–v0.9.34 (29 versions) exhaustively tested the Sturmian bond sphere for LiH. Every variant failed. This section distills the failure into a single structural theorem that explains why the Sturmian basis, despite its geometric elegance, cannot produce geometry-dependent binding in the shared-$p_0$ framework without solving a continuous PDE at every $R$.

**Theorem 4** (Sturmian structural theorem)**.** *In the molecular Sturmian basis with shared momentum scale $p_0$, the one-electron Hamiltonian matrix satisfies*

$$H_{ij} = \frac{p_0^2}{2}\left(\frac{1}{\beta_j} - 1\right) S_{ij}, \tag{34}$$

*where $S_{ij}$ is the overlap matrix and $\beta_j$ is the Sturmian eigenvalue of orbital $j$. The Hamiltonian is proportional to the overlap matrix. Consequently, the generalized eigenvalues*

$$E_k = \frac{p_0^2}{2}\left(\frac{1}{\beta_k} - 1\right) \tag{35}$$

*are independent of the bond angle $\gamma$ and hence independent of the internuclear distance $R$.*

*Proof.* Each Sturmian function $S_j$ satisfies $(-\frac{1}{2}\nabla^2 - \beta_j V_{\mathrm{mol}})S_j = E_0 S_j$ with $E_0 = -p_0^2/2$. The matrix element of the physical Hamiltonian $\hat{h}_1 = -\frac{1}{2}\nabla^2 + V_{\mathrm{mol}}$ between Sturmians $S_i$ and $S_j$ is

$$\begin{aligned} H_{ij} &= \langle S_i|\hat{h}_1|S_j\rangle \\ &= E_0\, S_{ij} + (1-\beta_j^{-1})\langle S_i|V_{\mathrm{mol}}|S_j\rangle. \end{aligned} \tag{36}$$

But $\langle S_i|V_{\mathrm{mol}}|S_j\rangle = -\beta_j^{-1}\langle S_i|(\hat{h}_1 - E_0)|S_j\rangle = -\beta_j^{-1}(H_{ij} - E_0 S_{ij})$, which gives the recursion. When $\beta_j$ is treated as a fixed parameter (independent of $R$), the overlap matrix $S_{ij}$ depends on $\gamma(R)$ through the $D$-matrix coupling, but $H_{ij}/S_{ij}$ does not. The generalized eigenvalue problem $H\mathbf{c} = E\,S\mathbf{c}$ then yields eigenvalues $E_k$ that depend only on $\{p_0, \beta_k\}$, not on $\gamma$. The nuclear repulsion

$V_{NN} = Z_A Z_B/R$ adds a purely $1/R$ potential, producing a monotonically repulsive total energy with no minimum. This was confirmed numerically in v0.9.34 diagnostic (bond sphere Hamiltonian analysis): $E_{\text{total}}(R)$ is a pure $1/R$ curve with no bound state. □

**Corollary 2** (Binding requires $\beta_k(R)$)**.** *Binding in the Sturmian framework requires the $R$-dependent prolate spheroidal eigenvalue spectrum $\beta_k(R)$. This in turn requires solving the two-center Schrödinger PDE at every $R$, which reintroduces continuous spatial geometry and violates the discrete graph premise of GeoVac.*

**Corollary 3** (Heteronuclear dual-$p_0$ failure)**.** *For heteronuclear diatomics, no shared $p_0$ exists that simultaneously represents both atomic length scales. The dual-$p_0$ extension (v0.9.34) fails because molecular Sturmian functions are inherently delocalized over both centers: at $p_{0,H} = 1.0$, the H-scale $1\sigma'$ orbital "sees" the Li nucleus ($Z_A = 3$) and acquires $H_1 = -2.6$ Ha. The self-consistent loop drives both $p_0$ values toward a single geometric mean ($p_0 \approx 2.0$), eliminating the scale separation that was the motivation for the extension.*

## XIII. $SO(4)$ SELECTION RULES FOR $\sigma$-BOND FORMATION

The $SO(4)$ Wigner $D$-matrix encodes the full angular coupling between atomic orbitals under the bond rotation $e^{i\gamma A_y}$. Two diagonal elements exhibit remarkable constancy.

**Theorem 5** ($\sigma$-bond selection rules)**.** *For the $n = 2$ representation of $SO(4)$:*

$$D^{(2)}_{(1,0),(1,0)}(\gamma) = 1 \quad \textit{for all } \gamma, \tag{37}$$

$$D^{(2)}_{(0,0),(1,0)}(\gamma) = 0 \quad \textit{for all } \gamma. \tag{38}$$

*That is, the bond rotation generated by $A_y$:*

1. *preserves the $2p_0$ orbital ($l = 1, m = 0$) without any attenuation at any bond angle, and*
2. *does not mix the $2s$ orbital ($l = 0, m = 0$) with the $2p_0$ orbital ($l = 1, m = 0$).*

*Proof.* The $D$-matrix in the $(l, m)$ basis is computed from the $\mathbf{J}^{(\pm)}$ Clebsch–Gordan decomposition (Eq. (**??**)) with $j = (n-1)/2 = 1/2$ for $n = 2$. The opposite-sign rotation $d^{1/2}(+\gamma) \otimes d^{1/2}(-\gamma)$ yields, for the $(l' = 1, m' = 0) \to (l = 1, m = 0)$ element:

$$D^{(2)}_{(1,0),(1,0)} = \cos^2(\gamma/2) + \sin^2(\gamma/2) = 1. \tag{39}$$

For the $(l' = 0, m' = 0) \to (l = 1, m = 0)$ element:

$$D^{(2)}_{(0,0),(1,0)} = \cos(\gamma/2)\sin(\gamma/2) - \sin(\gamma/2)\cos(\gamma/2) = 0. \tag{40}$$

Both identities hold for all $\gamma \in [0, \pi]$. □

*a. Physical interpretation.* The first identity states that a $2p_0$ ($\sigma$) orbital is perfectly transmitted through the bond rotation at any internuclear distance—it is a *transparent mode* of the $SO(4)$ bond channel. The second states that the $2s$ orbital cannot scatter into $2p_0$ via the bond rotation alone. Together, these enforce the quantum mechanical selection rule that $\sigma$-bonds are formed by $p_z$ orbitals aligned along the bond axis, not by $s$–$p$ mixing through the Runge–Lenz rotation. The $s$–$p$ hybridization that creates $\sigma$-bonds in the conventional picture arises from the *energy optimization* (variational principle), not from the geometric coupling.

*b. Numerical verification.* These identities were verified numerically across 10,000 values of $\gamma \in [0.01, \pi - 0.01]$ using the `wigner_D_so4` implementation (v0.9.15). The computed values satisfy $|D^{(2)}_{(1,0),(1,0)}(\gamma) - 1| < 10^{-14}$ and $|D^{(2)}_{(0,0),(1,0)}(\gamma)| < 10^{-14}$ at every grid point (`debug/test_harmonic_phase_lock.py`).

## XIV. CONCLUSION

This paper establishes two classes of results for the bond sphere—the description of a diatomic molecule as a single $S^3$ with two weighted poles.

*a. Positive structural results.* The bond sphere geometry is consistent and theoretically motivated. The internuclear distance encodes as a polar angle $\gamma$ via $\cos\gamma = (p_0^2 - p_R^2)/(p_0^2 + p_R^2)$, and all cross-atom matrix elements are $SO(4)$ Wigner $D$-matrix elements evaluated at the bond rotation $e^{i\gamma A_y}$ (Eq. (**??**)). The Runge–Lenz alignment argument (Sec. **??**) shows that $s$–$p$ hybridization is a dynamical consequence of the two-center potential. For heteronuclear molecules, the bond sphere becomes an egg—a prolate spheroid whose asymmetry encodes bond polarity (Sec. **??**). The kinetic membrane (Sec. **??**) provides a discrete analog of Bader's bond critical point. The $SO(4)$ selection rules (Theorem **??**) show that $D^2_{10,10}(\gamma) = 1$ and $D^2_{00,10}(\gamma) = 0$ for all $\gamma$: the $2p_0$ orbital is a transparent mode of the bond channel, and $s$–$p$ mixing requires the variational principle, not the geometric coupling. The single-atom lattice of Papers 0–7 is the $\gamma = 0$ limit; the 18 symbolic proofs and the universal constant $-1/16$ are unchanged.

*b. Negative theorem: Sturmian basis limits.* The Sturmian structural theorem (Theorem **??**) proves that in the shared-$p_0$ Sturmian basis, the one-electron Hamiltonian is proportional to the overlap matrix, making eigenvalues independent of the bond angle $\gamma$ and hence of $R$. Binding requires the full $R$-dependent prolate spheroidal eigenvalue spectrum $\beta_k(R)$, which reintroduces continuous spatial geometry (Corollary **??**). For heteronuclear diatomics, no shared $p_0$ simultaneously represents both atomic length scales (Corollary **??**). This theorem was verified computationally across 29 diagnostic versions (v0.9.20–v0.9.34), all of which failed for LiH by the predicted mechanism.

*c. Computational path forward.* The bond sphere is geometrically consistent but its computational implementation requires either the full LCAO architecture—which preserves atom-specific length scales and produces $D_e^{\mathrm{CP}} = 0.093$ Ha at nmax= 3 (1.0% error vs. experiment, documented in the companion paper [**?** ])—or a non-Sturmian basis that avoids the proportionality constraint of Theorem **??**. The LCAO path is validated; the bond sphere geometry provides the theoretical framework within which the LCAO results are understood.

*d. Resolution: the prolate spheroidal lattice.* The failure of the single-$p_0$ Bond Sphere approach points toward the correct solution: rather than forcing both centers onto a shared $S^3$ with one momentum parameter, each center retains its identity while being embedded in the natural two-center geometry—the prolate spheroid. In prolate spheroidal coordinates $(\xi, \eta, \varphi)$, the two-center Coulomb problem separates exactly, and discretizing the Laplace-Beltrami operator on this manifold yields a molecular lattice with $R$-dependence built into the metric. For $H_2^+$, this approach achieves $R_{\mathrm{eq}} = 2.001$ bohr (0.21% error) and $E_{\min} = -0.598$ Ha (0.70% error) with zero fitted parameters [**?** ].

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